

# 08 Zerotrust

## Zero-Trust Local Mesh Benchmark — CASPER

**Company:** Anticloud FZ LLE

### Certificate Operations

Operation Time

--- ---

CA generation (4096-bit RSA) 1.8s (one-time)

Service cert issuance (2048-bit) 0.4s

mTLS handshake 2.1ms

Cert fingerprint (SHA-256) 0.1ms

### Security Properties

Property Value

--- ---

CA key size 4096-bit RSA

Service cert key size 2048-bit RSA

Service cert TTL 1 day (configurable)

TLS version 1.3 (minimum)

Cipher suites TLS\_AES\_256\_GCM\_SHA384, TLS\_CHACHA20\_POLY1305\_SHA256

mTLS enforcement Both client and server certs required

### Attack Resistance

Attack Mitigation Status

--- --- ---

MITM on localhost mTLS — attacker cannot present valid cert MITIGATED

Stolen service cert TTL 1 day — rotates automatically MITIGATED

CA key compromise CA key never transmitted, air-gapped MITIGATED

Downgrade to TLS 1.2 Minimum TLS 1.3 enforced MITIGATED